

Photonic Millimeter-wave Parallel Chaos Source Enabling Multi-Channel Secure Communication and Sensing

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Abstract We experimentally demonstrate a 2×5 Gbps photonic mmWave secure communication and an interference-free MISO sensing with a single receiver, empowered by a parallel chaos source with 40 channels and up to 94.5 GHz bandwidth per channel, overcoming hardware bottlenecks in future multi-functional mmWave networks. ©2026 The Author(s)

Introduction

Millimeter-wave (mmWave) networks for high-speed communication and high-resolution sensing are severely limited by physical-layer security vulnerabilities and interference [1–3]. Photonic chaos has emerged as a powerful solution due to its broadband and noise-like characteristics, which naturally provide secure encryption [4–7] and robust anti-interference sensing [8–11]. To support complex network scenarios, generating multiple chaotic signals in parallel is essential for scaling system capacity and enabling multi-functional operations.

However, scaling these systems remains major bottlenecks. To support multiple independent channels, conventional schemes typically rely on duplicating discrete laser sources and electro-optic comb components [12–14], leading to prohibitive hardware costs for parallelization. Alternatively, when employing frequency division multiplexing from a single broad source, there exists a fundamental trade-off between the number of independent channels and the achievable single-channel

bandwidth [15–18], limiting its applications for high capacity and high accuracy. Furthermore, separating spatially overlapped signals conventionally necessitates bulky multi-antenna arrays, otherwise suffering from severe inter-channel crosstalk [19,20]. Achieving scalable and hardware-efficient parallel mmWave source without crosstalk is a critical task for future mmWave networks.

In this paper, we propose a photonic mmWave parallel chaos source to enable secure multi-channel communication and interference-free multi-input single-output (MISO) sensing. Enabled by the integrated Mach Zehnder interferometer (MZI) mesh, we achieve 40 independent channels with an ultra-broad bandwidth of up to 94.5 GHz per channel, depicted in Fig. 1. We demonstrate 2×5 Gbps secure access and successfully resolve multiple targets from mixed mmWave echoes using merely a single receiver. This architecture provides a highly scalable and efficient solution for mmWave integrated sensing and communication (ISAC) networks.

Architecture and Experimental Setup

The conceptual architecture of the parallel mmWave chaos transmitter driven by a spatially parallel photonic chaos source is illustrated in Fig. 2 [21]. The core of the transmitter is an optoelectronic feedback loop incorporating a photonic integrated MZI mesh. The mesh acts as a critical phase-to-intensity converter, providing essential nonlinearity for chaotic oscillations. Crucially, the independent control and inherent fabrication tolerances of each MZI node ensure strictly distinct nonlinear transformations along each spatial path, yielding highly de-correlated, mutually independent parallel chaotic outputs. A portion of the mesh output is fed back to the phase modulator (PM) to sustain the oscillations. Finally, these baseband chaotic carriers are optically heterodyned with

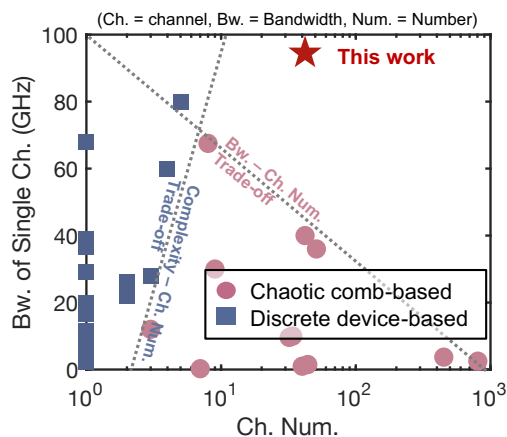


Fig. 1: Representative demonstrations of parallel chaos source.

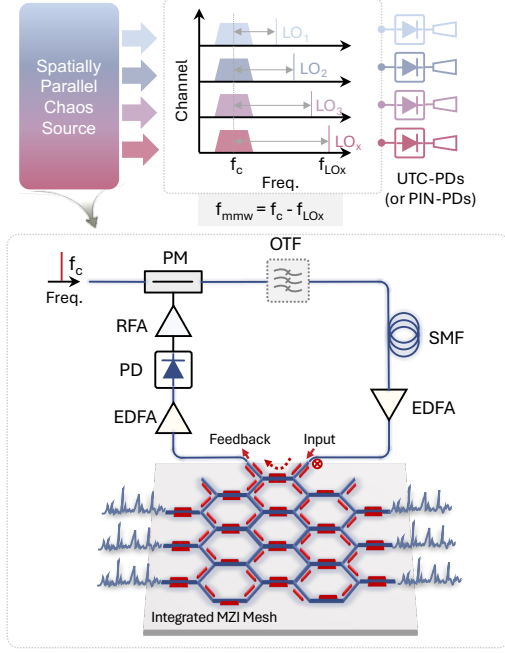


Fig. 2: Conceptual architecture of the parallel mmWave chaos transmitter driven by a spatially parallel photonic chaos source.

distinct local oscillators (LOs) at uni-traveling-carrier photodiodes (UTC-PDs) to generate independent mmWave bands ($f_{mmw} = f_c - f_{LOx}$).

Fig. 3 shows the detailed experimental setup for multi-user secure access and MISO sensing. A seed laser is split to drive both the chaos generation and message modulation modules. Two 5 Gbaud NRZ messages are encoded via Mach Zehnder modulators (MZMs) and optically coupled with parallel chaotic signals for physical-layer masking. Concurrently, an optical frequency comb (OFC) provides two specific tones via a WSS to serve as LOs. The masked signals and LOs are combined, amplified, and injected into UTC-PDs, where optical heterodyning up-converts them to mmWave bands for free-space emission via horn antennas. The system

operates in two distinct configurations, shown in the upper right insert of Fig. 3. For the setup of multi-user secure access, it functions as a 2×2 link for User 1 and User 2. For the setup of MISO sensing, the communication message branches are strictly attenuated to zero via the VOAs. Two pure chaotic signals from two independent channels (CH1 and CH2) illuminate Target 1 and Target 2, respectively, which are physically placed with a relative distance of 10 cm. Crucially, the reflected mixed echoes are collected by merely a single receiver (Rx). In both scenarios, the setup for Rx utilizes an electrical mixer equipped with a 12× frequency multiplier chain to down-convert the captured mmWave signals to an intermediate frequency (IF), which is then digitized by an oscilloscope (OSC) for offline DSP.

Experimental Results and Discussions

The spatially parallel photonic chaos source is firstly physically characterized, where 40 channels are available for chaos outputs. Fig. 4a displays the noise-like time-domain intensity waveform of a selected channel. The largest Lyapunov exponent (LLE) is calculated and remains positive across all 40 generated channels, confirming stable chaotic states rather than random noise, shown in Fig. 4b. Furthermore, the optical spectrum in the top of Fig. 4c and the bandwidth distribution in the bottom demonstrate that each independent channel exhibits an ultra-broad 10-dB optical bandwidth of up to 94.5 GHz, demonstrating the massive capacity of the underlying physical engine.

Fig. 4d displays the optical spectrum of the generated OFC. Two specific comb lines with a 15 GHz spacing are selected as the distinct optical LOs for User 1 and User 2, respectively. Fig. 4e illustrates the coupled optical spectrum before injecting into the UTC-PD. Notably, the two masked optical message signals share the

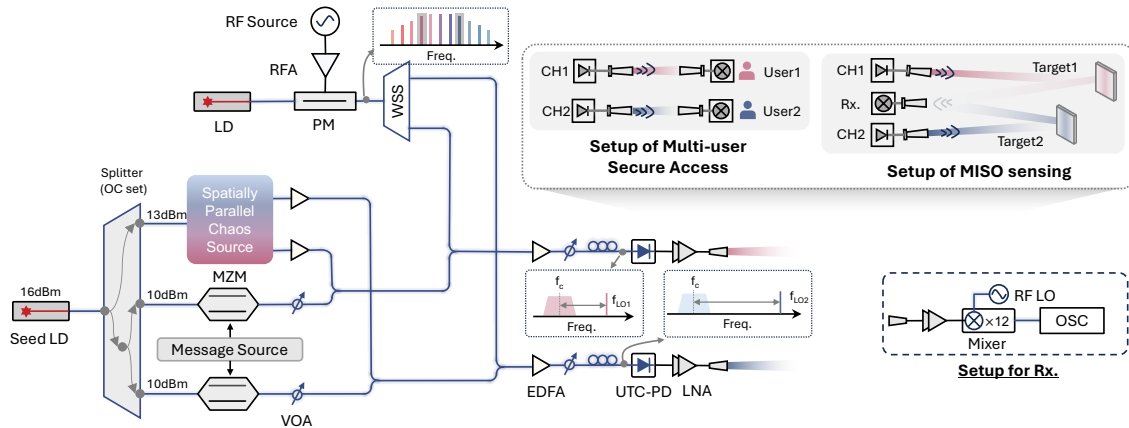


Fig. 3: Experimental setup for the multi-user secure access and MISO sensing system empowered by the spatially parallel photonic chaos source based on the integrated MZI mesh.

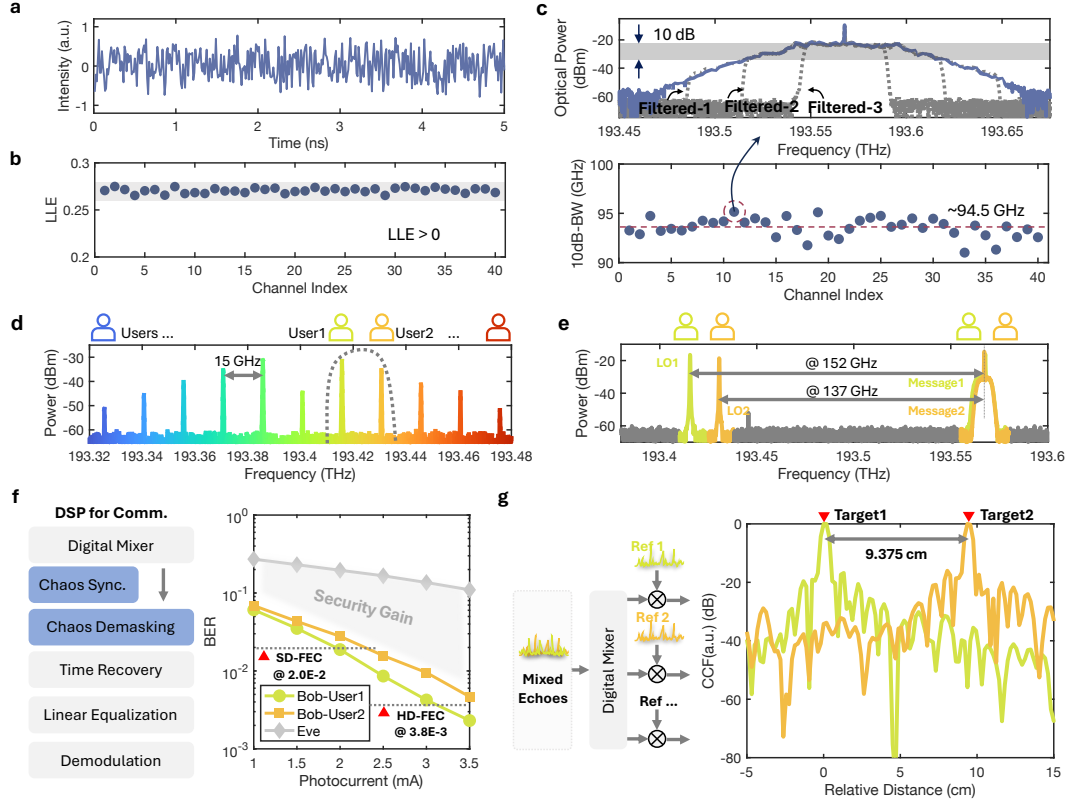


Fig. 4: Experimental results. (a) Time-domain waveform of generated chaos, (b) LLE across 40 channels, (c) Optical band-width across 40 channels. (d) Optical spectrum of OFC, (e) Optical spectrum of coupled signals for mmWave generation. (f) Communication performance in terms of BER, (g) Sensing performance in terms of CCF

same optical center frequency, whereas the two injected LOs are separated in frequency. Through optical heterodyne beating at the UTC-PD, the baseband signals are successfully up-converted to mmWave bands at 152 GHz and 137 GHz, respectively.

For multi-user secure access, the received IF signals of authorized users (Bob) are processed using the DSP routine in left insert of Fig. 4f. Specifically, the chaos synchronization is achieved via a neural network (NN) approach, as previously established by our previous work [22]. However, an unauthorized eavesdropper (Eve) can neither perfectly replicate the physical MZI-mesh architecture nor acquire the massive NN weight parameters, rendering chaos synchronization and demasking impossible. Consequently, Eve is forced to directly demodulate the masked signal. The right insert of Fig. 4f presents the measured bit error rate (BER) curves. As the received photocurrent increases, Eve's BER does decrease slightly due to improved overall signal power. However, the undemasked chaotic carrier acts as severe noise, keeping her BER above 0.1. In contrast, Bob successfully perform demasking, exhibiting a much steeper BER drop and achieving error-free recovery below the SD-FEC limit [23] and the HD-FEC threshold [24]. The performance gap visualizes the substantial physical-layer security

gain. Although the experiment is performed at 2×5 Gbaud, the 40-channel capacity of the underlying chaos source highlights the architecture's immense scalability.

Finally, for MISO sensing, the left insert of Fig. 4g outlines the target separation process, where the single overlapped IF echo is correlated with according local references. The resulting cross-correlation function (CCF) traces in the right insert of Fig. 4g accurately extract the distance peaks for Target 1 and Target 2. The measured relative distance of 9.375 cm aligns excellently with the 10 cm physical spacing. The absence of ghost targets rigorously proves that the strict independence of the MZI-mesh-generated chaos enables crosstalk-free multi-target sensing using a hardware-efficient single-receiver architecture.

Conclusions

We experimentally demonstrate a 2×5 Gbps photonic mmWave multi-channel secure communication transmission and an interference-free multi-target MISO sensing with merely a single receiver, empowered by a photonic spatially parallel chaos source with 40 independent channels and ultra-broad bandwidth of up to 94.5 GHz per channel. This hardware-efficient architecture provides a highly scalable solution for future secure ISAC networks, offering massive connectivity and sub-millimeter sensing potential.

Acknowledgements

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